R Programming Practical Examination

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Part A: Basic Operations

A1 Create v as the sequence 5, 10, 15, ..., 50 using `seq()`. Print its length, sum, and mean.

```
v <- seq(from = 5, to = 50, by = 5)

length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2 Create w that repeats the vector c(1, 2, 3) four times, and z that repeats each element of c(1, 2, 3) four times. Print both.

```
w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3 Compute and print: the remainder of 17 divided by 5, the integer division of 17 by 5, 2 to the power of 10, square root of 144, and pi rounded to 3 decimals.

```
17%/%5
```

```
## [1] 3
```

```r
2^10
```

```
## [1] 1024
```

```r
sqrt(144)
```

```
## [1] 12
```

```r
round(pi, 3)
```

```
## [1] 3.142
```

A4 Given a <- c(2, 4, 6, 8) and b <- c(1, 3, 5, 7), print a+b, a*b, the logical comparison a>b, and the dot product sum(a*b).

```r
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)
a+b
```

```
## [1]  3  7 11 15
```

```r
a*b
```

```
## [1]  2 12 30 56
```

```r
a>b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a*b)
```

```
## [1] 100
```

A5 Use paste() to produce "Student-1" "Student-2" "Student-3". Then identify the datatype of 42., 42L, "42", 41i, and TRUE. What are its output?

```r
paste("Student-", 1:3)
```

```
## [1] "Student- 1" "Student- 2" "Student- 3"
```

```r
typeof(42)
```

```
## [1] "double"
```

```r
typeof(42L)
```

```
## [1] "integer"
```

```r
typeof("42")
```

```
## [1] "character"
```

```r
typeof(41i)
```

```
## [1] "complex"
```

```r
typeof(TRUE)
```

```
## [1] "logical"
```

Part B: Data Frames and Conditional Statements

B1 Create a data frame students with 8 rows and these columns (use `stringAsFactors = FALSE`; sex is "M" or "F"):

```r
students <- data.frame(
  id = c(1, 2, 3, 4, 5, 6, 7, 8),
  name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farif", "Grace", "Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
  final = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)
students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farif   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

B2 Display the structure, dimensions, column names, and summary of students.

```r
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : num  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```r
dim(students)
```

```
## [1] 8 7
```

```r
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```r
summary(students)
```

```
##        id              name               sex             section
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3 Add a column `average` = the mean of midterm and final for each student, rounded to 1 decimal.

```r
students$average <- round((students$midterm + students$final) / 2, 1)
students$average
```

```
## [1] 80.0 82.0 94.0 69.0 77.0 89.5 94.0 62.0
```

B4 Add a logical column `passed` that is `TRUE` when`average >= 70` and `attendance >= 75`.

```r
students$passed <- (students$average >= 70 & students$attendance >= 75)
students$passed
```

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
```

B5 Using `ifelse()`, add a column `standing` with these values based on average:

- "Honors" if average >= 90
- "Good" if average >= 75 (and below 90)
- "Needs Support" otherwise

```r
students$standing <- ifelse(students$average >= 90, "Honors",
                            ifelse(students$average >= 75 & students$average < 90, "Good", "Needs Support"))
students$standing
```

```
## [1] "Good"          "Good"          "Honors"        "Needs Support"
## [5] "Good"          "Good"          "Honors"        "Needs Support"
```

B6 Write a function `classify(score)` that uses an `if / elseif / else` chain and returns `"Distinction"` for 85 or above, `"Merit"` for 70 to 84, `"Pass"` for 50 to 69, and `"Fail"` below 50. Test it with `sapply(c(90, 73, 55, 40), classify)` and print the result.

```r
classify <- function(score){
  if (score >= 85){
    return("Distinction")
  } else if (score >= 70 & score <= 84){
    return("Merit")
  } else if (score >= 50 & score <= 69){
    return("Pass")
  } else {
    return("Fail")
  }
}

sapply(c(90, 73, 55, 40), classify)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: `rbind()` and `cbind()`

C1 Create a data frame `new_students` with two students. Its columns must match the original, including `average`, `passed` and `standing`, computed the same way:

- id 9, Ines, F, section C, midterm 81, final 84, attendance 92
- id 10, Jamal, M, section A, midterm 70, final 68, attendance 80

Use `rbind()` to append them to `students`, saving the result as `students_all`.

```r
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 86),
  attendance = c(92, 80)
)

new_students$average <- (new_students$midterm + new_students$final) / 2
new_students$passed <- new_students$average >= 60
```

```r
new_students$standing <- ifelse(new_students$passed, "Good", "Needs Support")

students_all <- rbind(students, new_students)
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farif   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    86         80    78.0   TRUE          Good
```

C2 Create the numeric vector `quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)`. Use `cbind()` to attach it to `students_all` as a column named `quiz`. Confirm the result is still a data frame with `class()`.

```r
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)

students_all <- cbind(students_all)
```

C3 Create the matrix `m <- cbind(midterm = students_all$midterm, final = students_all$final)` and use `rbind()` to append a final row containing the column means, labelled "mean".

```r
m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
m
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  86.0
## mean    78.9  82.7
```

C4 In a comment (2-3 sentences), explain what error `rbind()` produces if the data frames have different column names, and show the output if you safely catch it with `tryCatch()`

```r
#rbind() requires data frames to have same column names and it produces an error.
```

Part D: Subsetting

Use `students_all` (with quiz) for all questions. Print every result.

D1 Extract the name column in three ways: `$`, `[["name"]]`, and `[ , "name"]`.

```
students_all$name
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farif" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farif" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[ , "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farif" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

D2 Extract rows 2 to 5 and only the columns `name`, `midterm`, `final`.

```
students_all[2:5, c("name", "midterm", "final")]
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

D3 Using logical subsetting, select students with `attendance` above 90.

```
students_all[students_all$attendance > 90, ]
```

```
##   id  name sex section midterm final attendance average passed standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good
```

D4 Using `subset()`, selendefinedndefinedct students of sex "F" in section "A" or "C" whose `final` is at least 75; return only `name`, `section` and `final`.

```
subset(students_all,
       sex == "F" & (section == "A" | section == "C") & final >= 75,
       select = c(name, section, final))
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

D5 Use `which.max()` and `which.min()` to report the name of the student with the highest and lowest `average`.

```
students_all$name[which.max(students_all$average)]
```

```
## [1] "Chen"
```

```
students_all$name[which.min(students_all$average)]
```

```
## [1] "Hiro"
```

D6 Sort `students_all` by `average` in descending order, then print the top 3 rows.

```
sorted_students <- students_all[order(-students_all$average), ]

head(sorted_students, 3)
```

```
##   id  name sex section midterm final attendance average passed standing
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors
## 6  6 Farif   M       C      88    91         85    89.5   TRUE     Good
```

Part E: Basic Factors

E1 Convert `section` and `sex` in `students_all` to factors with `factor()`. Print `levels()` of each.

```
students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)


levels(students_all$section)
```

```
## [1] "A" "B" "C"
```

```
levels(students_all$sex)
```

```
## [1] "F" "M"
```

E2 Use `table()` to count students per sex and per section.
*Try the two-way table of sex by section. `table(students_all$sex, students_all$section)`. What is the output?

```
table(students_all$sex)
```

```
##
## F M
## 5 5
```

```r
table(students_all$section)
```

```
##
## A B C
## 4 3 3
```

```r
table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

E3 Print `as.integer()` of the `sex` factor. In a comment, explain what the integers represent.

```r
as.integer(students_all$sex)
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

E4 Create a new factor `sex_label` from `sex` using `factor()` with `levels = c("F", "M")` and `labels = c("Female", "Male")`. Add it to `students_all` and print its `levels()`.

```r
students_all$sex_label <- factor(students_all$sex,
                                 levels = c("F", "M"),
                                 labels = c("Female", "Male"))

levels(students_all$sex_label)
```

```
## [1] "Female" "Male"
```

E5 Print `summary()` of `sex_label`. Then, using logical subsetting and `mean()`, print the mean average for each level of `sex_label`.

```r
summary(students_all$sex_label)
```

```
## Female   Male
##      5      5
```

```r
mean(students_all$average[students_all$sex_label == "Female"])
```

```
## [1] 80.5
```

```r
mean(students_all$average[students_all$sex_label == "Male"])
```

```
## [1] 81.1
```